Transcriptomic profile of drought-responsive pathways in *Oryza sativa*: a meta-analysis

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17th Bioinformatics and Computational Biology International Conference – BBCC2022

13-15 December 2022

A. Introduction

- Rice is a staple food that provides nutrition to more than half of the world's population.
- Drought is a major abiotic stress that can have a significant impact on rice crop yield.
- Identifying genes and gene products that are involved in drought response is critical [1].
- We investigated the impact of drought stress on two major rice subspecies, ssp. *Japonica* and ssp. *Indica* by conducting a meta-analysis of RNA-seq data.

B. Methods

- A systematic review was performed according to the PRISMA Statement in GEO database.
- Collection of all RNA-seq experiments regarding drought vs. normal seedling/shoot tissue.
- For consistency, all gene expression data were converted to TPM reads.
- Meta-analysis was conducted using MAGE [2] at a 0.5% level of significance (FDR).
- Protein-Protein Interaction Network construction using STRING online database tool for all differentially expressed proteins.
- STRING, g:Profiler, and PANTHER databases were utilized for Pathway Enrichment Analysis.

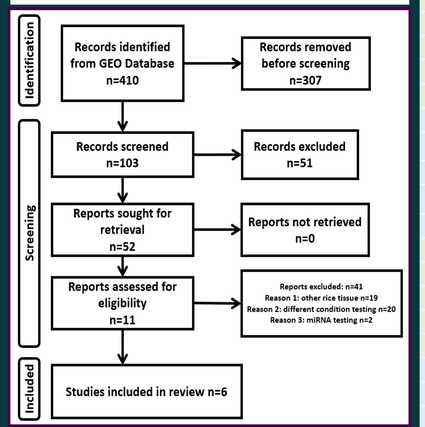

Figure 1. PRISMA flow diagram representing the stages during the systematic reviewing process for study collection.

C. Results

- Six studies out of 410 were considered eligible for meta-analysis.
- 1787 differentially expressed genes (DEGs) were initially identified in seedling/shoot tissues at a 0.5% significance level (FDR), including 877 upregulated and 910 downregulated genes.
- A protein-protein interaction (PPI) network was established using STRING where 1398 out of 1787 DEGs, with a confidence score ≥0.7, were recognized.
- The initial network was further investigated in terms of the experimental interactions that the proteins exhibit with each other.
- Top six nodes, namely *Os02g0281000, Os01g0896700, Os04g0462500, Os05g0430400, Os05g0557200, Os08g0412200,* were identified as hub genes and their function was further examined.

Table 1. Summary results of analysis and meta-analysis of six individual studies using MAGE tool (FDR 0,005).

Study	Author	Public on Year	Platform	Tissue	Sample size (control/case)	Deferentially Expressed Genes
GSE81462	Zong W et al.	2016	GPL13834 Illumina HiSeq 2000	Seedling	1/1	38247
GSE117047	Ma S et al.	2019	GPL13160 Illumina HiSeq 2000	Seedling	3/6	9433
GSE128495	Li X et al.	2019	GPL23876HiSeq X Ten	Seedling	3/3	5739
GSE160679	Yin L et al.	2021	GPL28757 Illumina HiSeq 3000	Seedling	3/2	17659
GSE147158	Sevanthi AM et al.	2020	GPL25149 Illumina HiSeq 1500	Shoot	2/2	4385
GSE107425	Fu J et al.	2018	GPL13834 Illumina HiSeq 2000	Shoot	3/2	8572
Meta-analysis of all six studies						**1787**

Table 2. Pathway Enrichment Analysis Results. Top five pathways according to GO:MF-Molecular function, GO:BP-Biological Process, GO:CC-Cellular Component, KEGG.

	Term	Term ID	adjusted p-value
GO:MF	phosphoric ester hydrolase activity	GO:0042578	0.008999673
GO:MF	catalytic activity	GO:0003824	0.010054542
GO:MF	nucleotidase activity	GO:0008252	0.018526027
GO:MF	phosphotransferase activity, alcohol group as acceptor	GO:0016773	0.029623622
GO:MF	kinase activity	GO:0016301	0.032400608
GO:BP	photosynthesis	GO:0015979	5.16E-07
GO:BP	generation of precursor metabolites and energy	GO:0006091	2.08076E-05
GO:BP	small molecule metabolic process	GO:0044281	0.000158216
GO:BP	phosphate-containing compound metabolic process	GO:0006796	0.00794996
GO:BP	phosphorus metabolic process	GO:0006793	0.010149127
GO:CC	chloroplast	GO:0009507	4.96E-27
GO:CC	plastid	GO:0009536	1.39E-24
GO:CC	plastid stroma	GO:0009532	1.52E-17
GO:CC	chloroplast stroma	GO:0009570	9.80E-17
GO:CC	plastid envelope	GO:0009526	3.69E-10
KEGG	Glyoxylate and dicarboxylate metabolism	KEGG:00630	4.13E-08
KEGG	Carbon metabolism	KEGG:01200	0.000361659
KEGG	Metabolic pathways	KEGG:01100	0.000538531
KEGG	Carbon fixation in photosynthetic organisms	KEGG:00710	0.002185149
KEGG	Photosynthesis	KEGG:00195	0.009355749

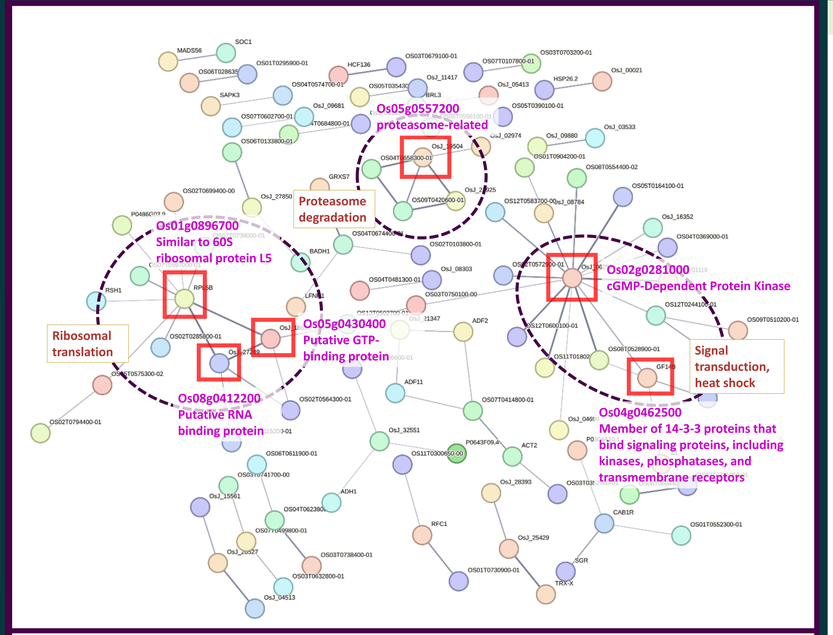

Figure 3. STRING Protein-Protein Interaction Network corresponding to seedling-shoot tissue DEGs. The network consists of 105 nodes at a 0.4 interaction score. Edges indicate experimentally determined P-P interactions. Red squares highlight the top six highly connected nodes.

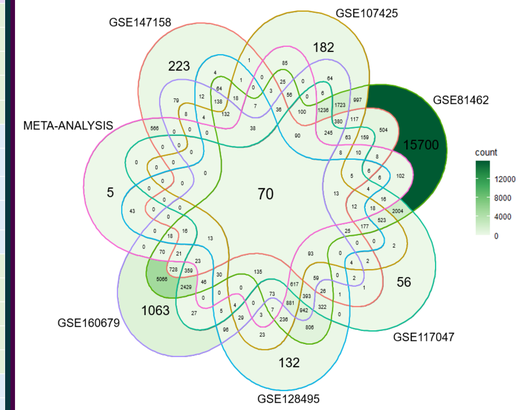

Figure 2. Venn diagram illustrating the DEGs between the meta-analysis and the six individual studies which we retrieved from GEO database. In green contrast are DEGs discovered through the meta-analysis but not in any of the primary investigations.

D. Conclusions

- Meta-analysis of RNA-seq data revealed **1787 DEGs** (differentially expressed genes) in rice under drought compared to control (877 upregulated and 910 downregulated).
- Enrichment analysis in STRING, g:Profiler and PANTHER databases for Gene Ontology terms and KEGG pathways revealed that these 1787 gene products are involved in **carbon and carbohydrate metabolism, photosynthesis, phosphorus metabolic process**, have **kinase, phosphotransferase** and other catalytic activities, and are located in **chloroplast/plastid membranes and stroma**.
- STRING **Protein-Protein Interaction** (PPI) **Network** analysis revealed 6 hubs (highly connected gene products).
- The six hub genes present divergent functions such as protein-serine / threonine **phosphatase and kinase activity**, are involved **in protein translation and degradation** and are putative **GTP- and RNA-binding** proteins with further **unknown functions**.
- Gene products involved in protein metabolism, signal transduction, and heat stress subnetworks are downregulated, while proteins in the proteasome subnetwork are overexpressed, implying that apoptosis may regulate response to drought stress.

Table 3. List of 6 hub genes and their protein functions. The arrow (↑↓) indicates up/down-regulation

Gene name	Protein name; Function; Subcellular Location
Os02g0281000 (↓)	OsJ_06259 - Protein-serine/threonine phosphatase; Hydrolase, protein phosphatase; cAMP-dependent protein kinase complex, cGMP-Dependent Protein Kinase
Os04g0462500 (↓)	GF14B (14-3-3-like protein GF14-B); member of 14-3-3 proteins that bind signaling proteins, including kinases, phosphatases, and transmembrane receptors; cytoplasm, nucleus
Os01g0896700 (↓)	RPL5B (60S ribosomal protein L5-2); Component of the ribosome, a large ribonucleoprotein complex responsible for the synthesis of proteins in the cell; cytoplasm, nucleus
Os05g0430400 (↓)	OsJ_18635; Putative GTP-binding protein; unknown function; mitochondrion
Os08g0412200 (↑)	OsJ_27289; Putative RNA binding protein; unknown function
Os05g0557200 (↑)	OsJ_19504; proteasome-related; unknown function

D. Acknowledgements and Funding

We acknowledge support by the projects: 1. "GENOMIC-OASIS: GENOMIC Analysis of Organisms of Agricultural and livestock Interest in Sterea", Grant Number (MIS) 5045902.The project IntDaMuS (Code 3198) is implemented under the "ARISTEIA II" Action of the "OPERATIONAL PROGRAMME EDUCATION AND LIFELONG LEARNING" and is co-funded by the European Social Fund (ESF) and National Resources.

2. Operational Program "STEREA ELLADA" 2014 - 2020 (Strengthening of Research Projects, Technological Development and Innovation in the fields of RIS3) "Molecular identification and exploitation of indigenous hop varieties for the production of high-quality beers with designation of origin-PAZL (Production Analysis of Hop Beers)-STER1-0026139".

D. References

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